

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 2 and 3 (Ch. 2)

The Macro Indicators

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AGENDA

- The essential macro indicators
- Definition
- Computation
- Limitations
- Relation among them

If you were the Prime Minister of your country, what characteristics would you want your economy to have?

1. ...

2. ...

3. ...

THE BASIC MACRO INDICATORS

The main indicators used to describe an economy

- **GDP**: production and income
- **Inflation rate**: price dynamics
- **Unemployment rate**: people looking for a job in the formal labor market

Many other are useful:

- Import , export
- Public deficit
- Interest rate
- Saving
- ...



*What is their meaning?
How are they measured?
What do they measure?*

WHY DO POLITICIANS AND ANALYSTS MONITOR THESE INDICATORS?

- Assess economic health
- Take decisions
- Predict and prevent recessions
- Manage inflation risks
- Support business and investment planning
- Evaluate social and economic progress
-

Euro zone economy stagnated last quarter, foreshadowing weak 2025

30 January 2025
Reuters

By Balazs Koranyi

January 30, 2025 11:05 AM GMT+1 · Updated 6 days ago



FRANKFURT, Jan 30 (Reuters) - The euro zone economy **stagnated last quarter** as worried consumers zipped up their purses, adding to fears that a long-predicted recovery could be further delayed, Eurostat data showed on Thursday.

Gross domestic product in the 20 nations sharing the euro was unchanged compared with the previous quarter, falling short of expectations for a 0.1% expansion in a Reuters poll, as two straight years of contraction in Germany weighed on the bloc as a whole.

Euro zone growth has been anaemic for the past two years as industry is in deep recession on high energy costs, governments have little cash to spend and households have saved more, hurting consumption. This weakening trend has only accelerated in recent months on **concern that the labour market is softening** and a **trade war with the United States could drag an already weak economy even lower**.

Part of the quarterly growth miss may be down to **Ireland**, however, which **recorded a 1.3% decline**. The heavy presence of multinationals there, including the world's top tech and pharmaceutical companies, distorts its growth figures and causes volatility.

Compared with a year earlier, fourth-quarter growth in the euro zone was up 0.9%, below expectations for 1.0%. Among the euro zone's biggest economies, **Germany and France both contracted while Italy stagnated and Spain reported 0.8% growth**.

Growth is expected to accelerate through 2025 and rise in 2026 to the so-called "potential", or the steady state without stimulus and without generating excess inflation.

The problem is that this rate is only around 1.4%, well below the 1.8%-1.9% estimate for the U.S., suggesting that the euro zone will keep trailing the world's biggest economy for years to come.

A key reason for this is **that productivity growth in the euro zone is anaemic and structural faults**, from cumbersome regulation and political discord to fragmented markets, **will hold back any improvement**.

Preliminary estimate of GDP - Q4 2024

In the fourth quarter of 2024 the seasonally and calendar adjusted, chained volume measure of Gross Domestic Product (GDP) remained stable with respect to the previous quarter and increased by 0.5 per cent over the same quarter of previous year.

The fourth quarter of 2024 has had two days less with respect to the previous quarter and two more compared to the same quarter of previous year.

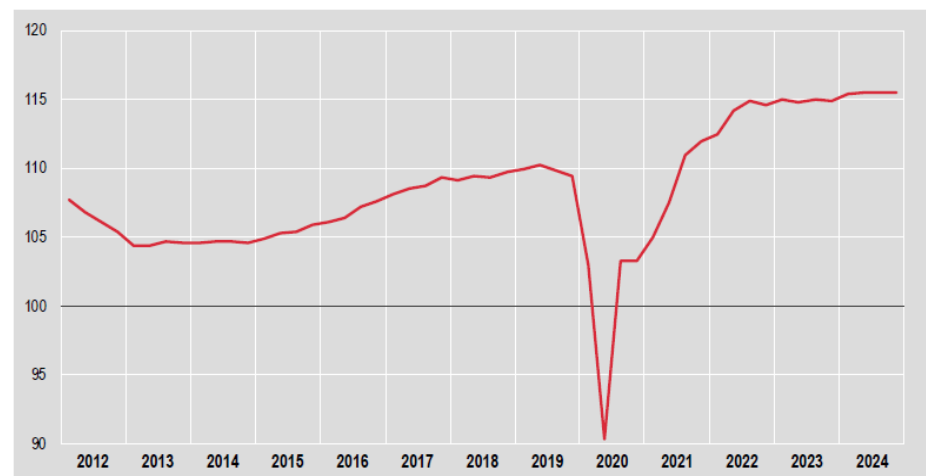
The quarter on quarter change is the result of a decrease of value added in both agriculture, forestry and fishing and in services, whereas the contribution of industry is positive.

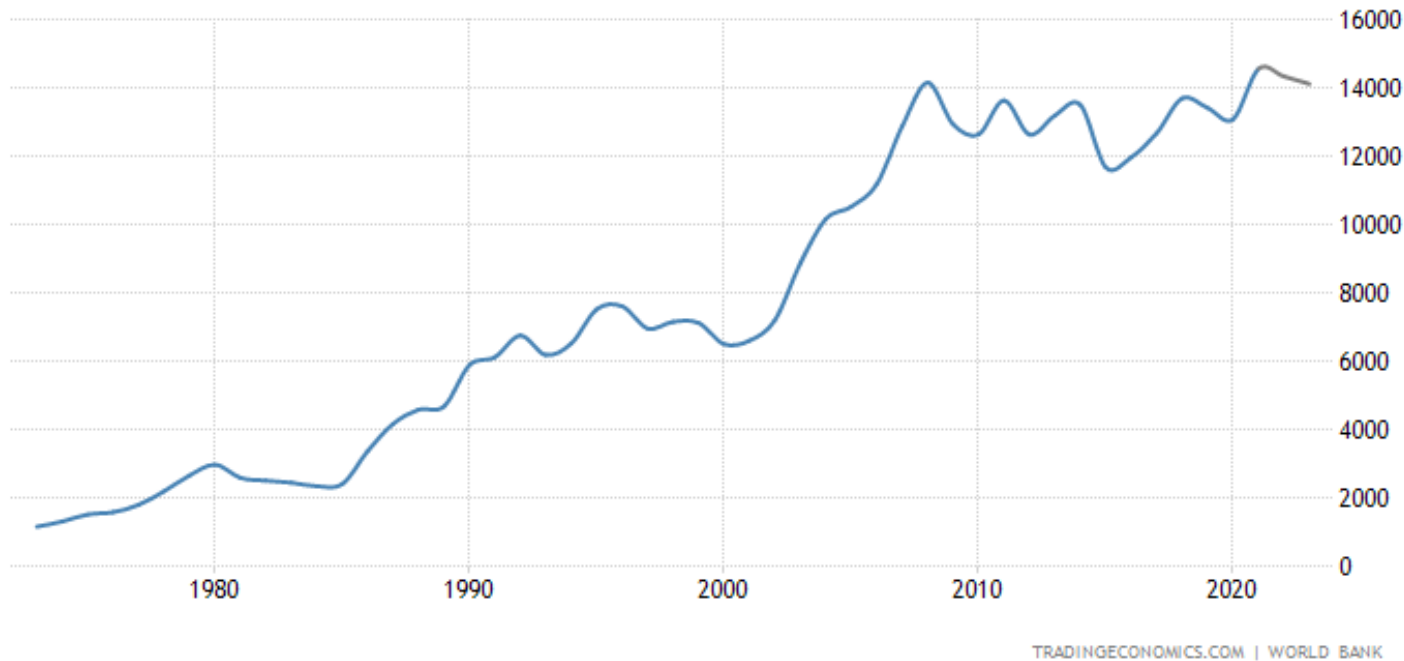
From the demand side, there is a negative contribution by the domestic component (gross of change in inventories) and a positive one by the net export component.

The carry-over annual GDP growth for 2025 is null.

<https://www.istat.it/en/press-release/preliminary-estimate-of-gdp-q4-2024/>

CHART 1. GROSS DOMESTIC PRODUCT, CHAIN-LINKED INDICES
Q1 2012 – Q4 2024, seasonally and calendar adjusted indices (reference year 2020)





AGGREGATE OUTPUT: GDP

Gross Domestic Product in the Euro Area (1960-2021) in USD Billion

<https://tradingeconomics.com/euro-area/gdp>

GROSS DOMESTIC PRODUCT – GDP



- Economy dimension
- It measures the aggregate output of an economy in a given period
- Constructed by the national income accounts systems
 - Define principles and rules
 - Collect data

A LITTLE BIT OF HISTORY

- No measures for aggregate economy in the 19th century or during the Great Depression
- Difficult to
 - to understand **what was happening** => put together bits and pieces of information (shipments, sales, reports...)
 - to **define interventions**
 - to **evaluate policies**



National income and product accounts

developed at the end of World War II to measure aggregate output on a regular basis (Kuznets and Stone)

Consider a simple economy with 2 firms

STEEL COMPANY		
Revenues from sales		100
Wages	80	
Profits		20

CAR COMPANY		
Revenues from sales		210
Wages	70	
Steel purchases	100	
Taxes	10	
Profits		30

How much is the GDP of this economy?

① VALUE OF PRODUCTION

GDP= sum of the value of all the final goods and services produced in the economy during a given period

- ❖ Value based on market prices
- ❖ Intermediate goods (goods used to produce other goods) excluded
- ❖ Only new productions, no second hand

Intermediate
good

Final good

= GDP =210

STEEL COMPANY		
Revenues from sales		100
Wages	80	
Profits		20

CAR COMPANY		
Revenues from sales		210
Wages	70	
Steel purchases	100	
Taxes	10	
Profits		30

② VALUE ADDED

GDP= sum of value added by all the firms in the economy during a given period

❖ VA = Value of production – Value of used intermediate goods

STEEL COMPANY		
Revenues from sales		100
Wages	80	
Profits		20

VA_1

$$GDP = VA_1 + VA_2 = 100 + 110 = 210$$

CAR COMPANY		
Revenues from sales		210
Wages	70	
Steel purchases	100	
Taxes	10	
Profits		30

$VA_2 = 210 - 100 = 110$

③ INCOME

GDP = sum of incomes generated in the economy during a given period

1. Labor income (wages) = W
2. Capital income (profits) = Π
3. Indirect taxes = T

STEEL COMPANY		
Revenues from sales		100
Wages	80	
Profits		20

CAR COMPANY		
Revenues from sales		210
Wages	70	
Steel purchases	100	
Taxes	10	
Profits		30

$$\text{GDP} = (80 + 70) + (20 + 30) + 10 = 210$$

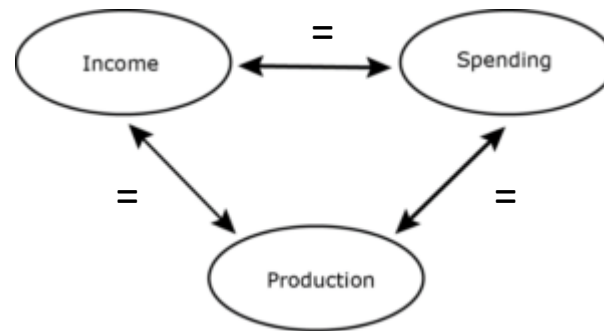
④ SPENDING

GDP= sum of expenditures for final goods and services in the economy during a given period

1. Consumer spending: C
2. Government spending on public goods and services: G
3. Business investment spending: I
4. Net exports = Exports – Imports : $NX = X - IM$

$$GDP = C + I + G + X - IM$$

SUMMING UP

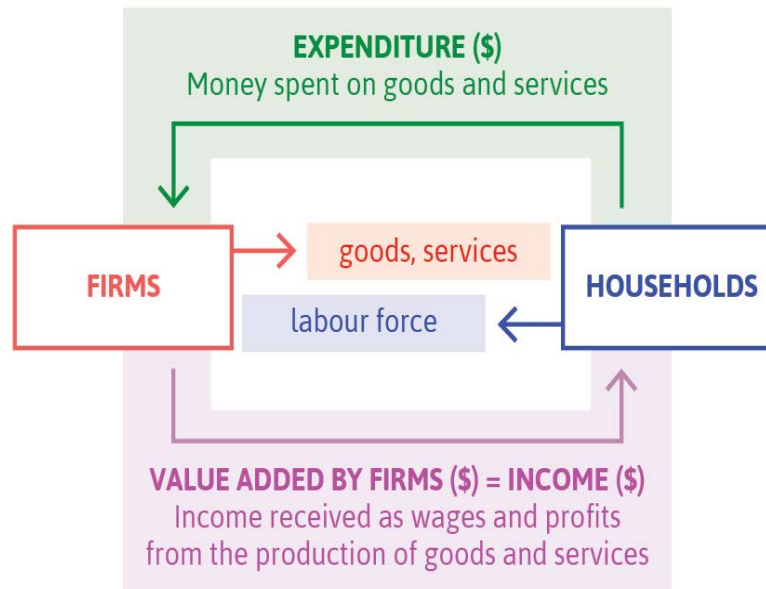


3 equivalent definitions and computation methods looking at:

1. Total domestic **production** in terms of
 - a. final goods and services produced in the economy
 - b. economy value added
2. Total domestic **income**
3. Total **spending** on domestic products

WHY DOES THE EQUIVALENCE BTW INCOME AND PRODUCTION HOLD?

- INTUITION: Income is used to buy final goods and services that are produced by someone else



Economic circular flow:
For each € produced/sold
there is
one € of income earned

WHY DOES THE EQUIVALENCE BTW INCOME AND VA HOLD?

Let's use our knowledge from microeconomics

$$\text{Profits} = \text{Revenues} - \text{Costs}$$

$$\text{Profits} = \text{Revenues} - \text{Costs for intermediate goods} - \text{Wages} - \text{Taxes}$$

$$\text{Profits} + \text{Wages} + \text{Taxes} = \text{Revenues} - \text{Costs for intermediate goods}$$

$$\text{Profits} + \text{Wages} + \text{Taxes} = \text{Value Added}$$

Incomes represent the way in which the value added is distributed in the economy



Value added is distributed among firms, workers, public sector!

WHAT IS IN AND WHAT IS OUT?

INCLUDED IN GDP

- Final goods and services produced by companies operating within the national territory, including
 - capital goods (machinery and plants)
 - newly constructed real estate
 - changes in inventories
- Exports
- Indirect taxes (VAT, tariffs, etc.)

NOT INCLUDED

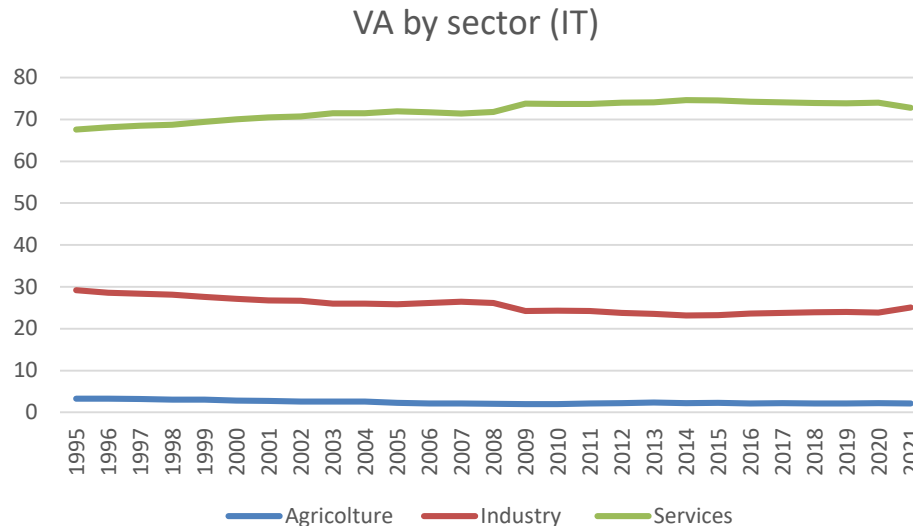
- Intermediate goods and services
- Used goods
- Financial activities (stocks, bonds, etc.)
- Goods and services produced abroad
- The production of non-market goods and services, such as "self-produced" goods and services (e.g., cooking and cleaning at home, taking care of children, etc.)
- Leisure and recreational activities
- The underground economy
- The production of illegal goods and services
- Pollution
- Criminality/corruption
- Impact of technology (free apps)
-

GDP OBSERVATIONS

1. GDP (gross Domestic product) : consider **where** production/income/spending occurs
 - *within the borders* of the country, even if produced by foreign factors of production!
2. GNP (Gross National product): consider the **nationality** of those producing/earning/spending
3. GDP is a **flow** measure=> production *during a given period*

LOOKING AT THE DATA

- VA produced by different sectors:
 - is stable in the short run
 - tends to vary in medium/long run => transition from less to more developed economies



- Stability in
 1. Income composition (about 40% is labor income, 45% capital income, 16% taxes)
 2. Types of spending (80% private and public consumption, 20% Investment)

IS GDP A GOOD MEASURE?

It depends!

What do we want to measure?

Well – being, Standards of living, Distribution...

[Kennedy, 18/3/1968](#)



REMARKS AT THE UNIVERSITY OF KANSAS, MARCH 18, 1968

Robert F. Kennedy
University of Kansas
March 18, 1968

Our Gross National Product, now, is over \$800 billion dollars a year, but that Gross National Product - if we judge the United States of America by that - counts air pollution and cigarette advertising, and ambulances to clear our highways of carnage. It counts special locks for our doors and the jails for the people who break them. It counts the destruction of the redwood and the loss of our natural wonder in chaotic sprawl. It counts napalm and counts nuclear warheads and armored cars for the police to fight the riots in our cities. It counts Whitman's rifle and Speck's knife, and the television programs which glorify violence in order to sell toys to our children.

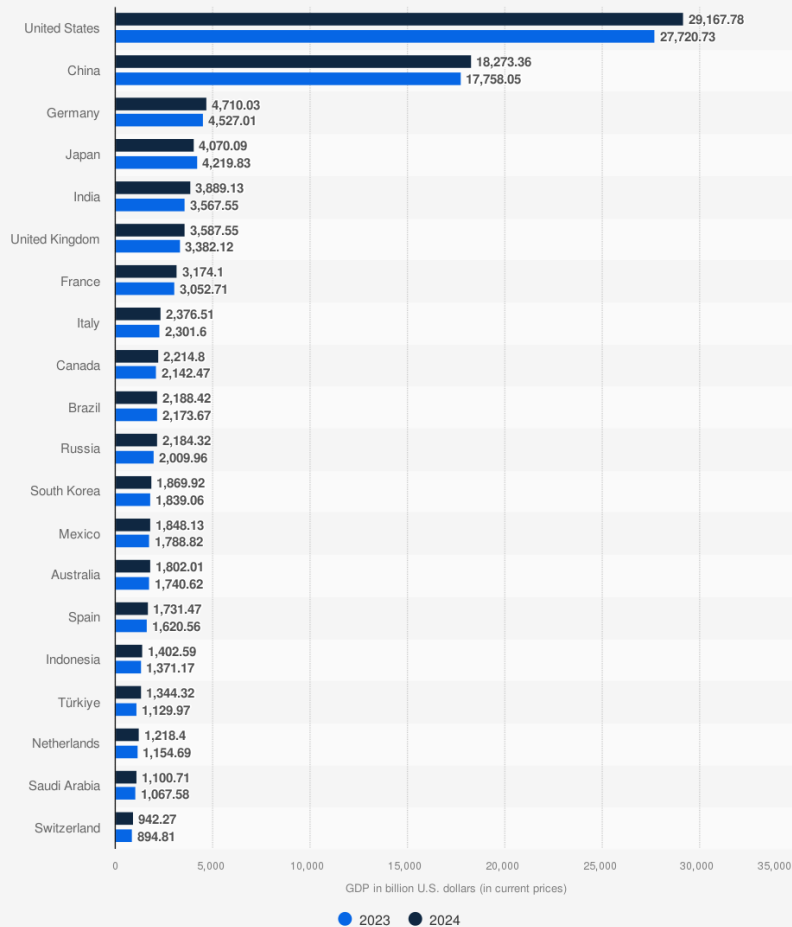
Yet the gross national product does not allow for the health of our children, the quality of their education or the joy of their play. It does not include the beauty of our poetry or the strength of our marriages, the intelligence of our public debate or the integrity of our public officials.

It measures neither our wit nor our courage, neither our wisdom nor our learning, neither our compassion nor our devotion to our country, it measures everything in short, except that which makes life worthwhile.

And it can tell us everything about America except why we are proud that we are Americans.

POPULATION SIZE AND GDP PER CAPITA (GDP/POPULATION SIZE)

The 20 countries with the largest gross domestic product (GDP) in 2024 (in billion U.S. dollars)

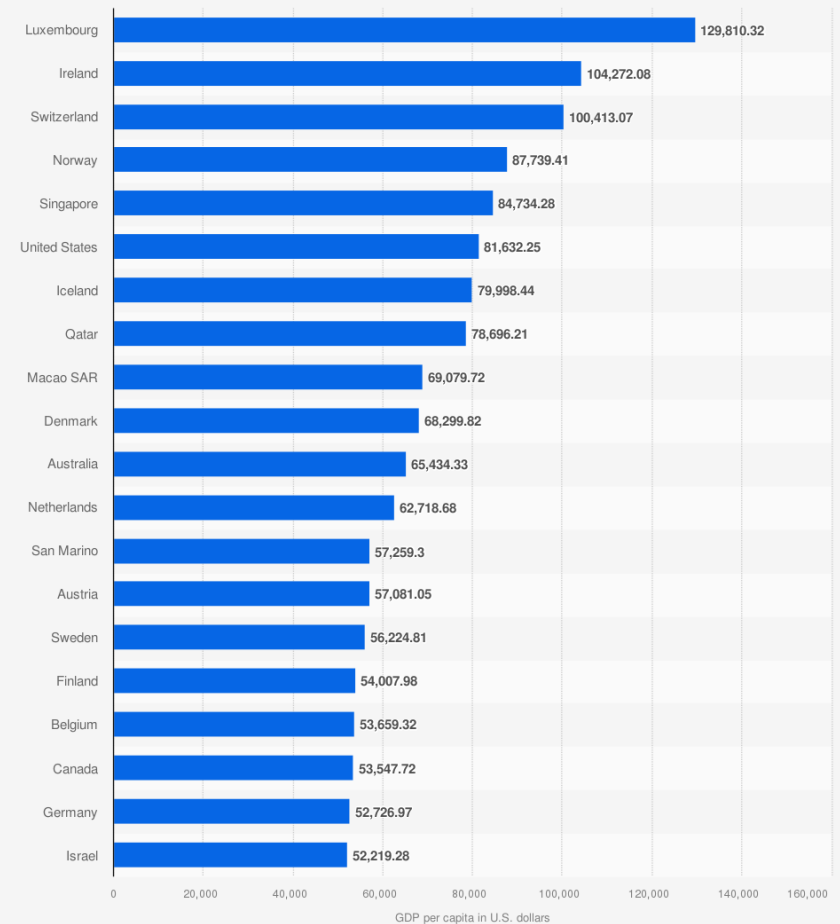


Source:
IMF
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Additional Information:
Worldwide; IMF

statista

The 20 countries with the largest gross domestic product (GDP) per capita in 2023 (in U.S. dollars)

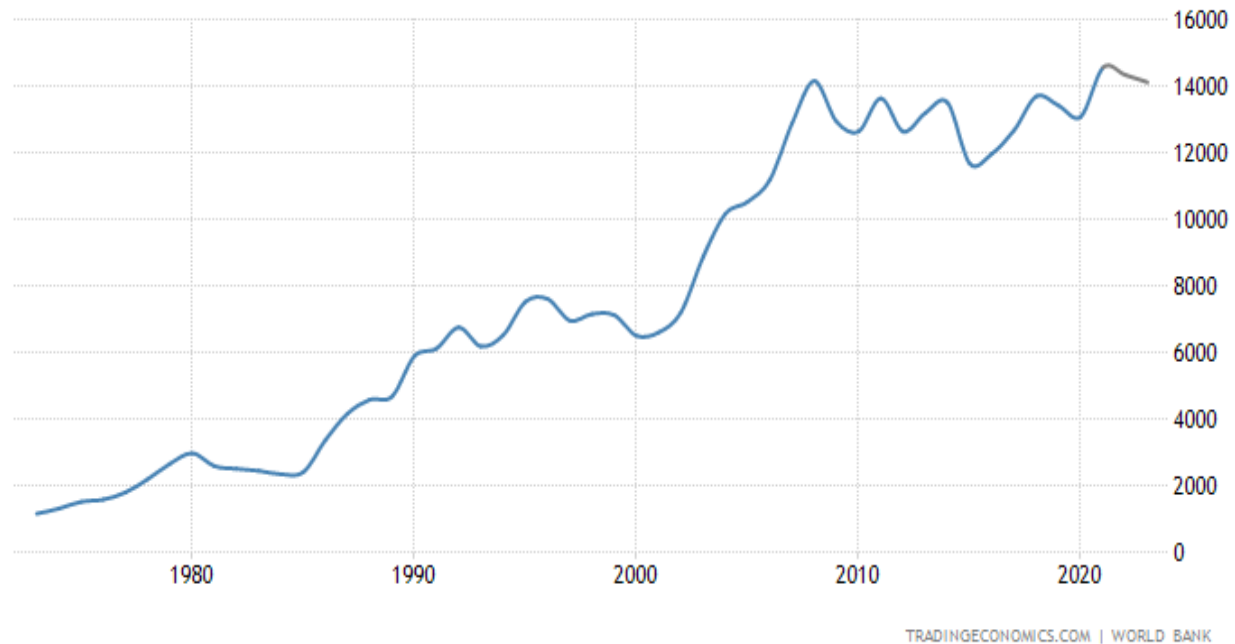


Source:
IMF
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Additional Information:
Worldwide; IMF

statista

***Are we richer now than 50
years ago?***



Gross Domestic Product in the Euro Area
(1960-2023) in USD Billion

Growth on divergent paths amid elevated policy uncertainty

Global growth is projected at 3.3 percent both in 2025 and 2026, below the historical (2000–19) average of 3.7 percent.

The forecast for 2025 is broadly unchanged from that in the October 2024 World Economic Outlook (WEO), primarily on account of an upward revision in the United States offsetting downward revisions in other major economies. Global headline inflation is expected to decline to 4.2 percent in 2025 and to 3.5 percent in 2026, converging back to target earlier in advanced economies than in emerging market and developing economies.

Medium-term risks to the baseline are tilted to the downside, while the near-term outlook is characterized by divergent risks. Upside risks could lift already-robust growth in the United States in the short run, whereas risks in other countries are on the downside amid elevated policy uncertainty. Policy-generated disruptions to the ongoing disinflation process could interrupt the pivot to easing monetary policy, with implications for fiscal sustainability and financial stability. **Managing these risks requires a keen policy focus on balancing trade-offs between inflation and real activity**, rebuilding buffers, and **lifting medium-term growth prospects** through stepped-up structural reforms as well as stronger multilateral rules and cooperation.

<https://www.imf.org/en/Publications/WEO/Issues/2025/01/17/world-economic-outlook-update-january-2025>

Value of production can be computed in
NOMINAL or REAL terms

NOMINAL GDP

- GDP is the value of the produced goods and services
 - Each good contributes with its price

Year	Quantity of Cars	Price of Cars
2011	10	\$20
2012	12	\$24
2013	13	\$26

Nominal GDP (at current prices) = value of current production based on prices of the period you are considering

$$€Y_t = p_{1t}Q_{1t} + p_{2t}Q_{2t} + \dots$$

Year	Quantity of Cars	Price of Cars	Nominal GDP
2011	10	\$20	\$200
2012	12	\$24	\$288
2013	13	\$26	\$338

Produced quantities ↑ Prices ↑

$$\begin{aligned}
 \text{Nominal GDP in 2011} &= 10 \text{ cars} \times \$20 \text{ per car} = \$200 \\
 \text{Nominal GDP in 2012} &= 12 \text{ cars} \times \$24 \text{ per car} = \$288 \\
 \text{Nominal GDP in 2013} &= 13 \text{ cars} \times \$26 \text{ per car} = \$338
 \end{aligned}$$

Changes do not depend only on real changes in the size of the economy

CHANGES IN NOMINAL GDP

◆ Nominal GDP changes for two reasons:

- Changes in the production
- Changes in prices

If our goal is to measure production and its change over time, we need to eliminate the effect of changes in prices (i.e. inflation/deflation)

How can we?

Necessary to construct another indicator that changes only if quantities change:

REAL GDP

REAL GDP

- We choose a **reference year (called base year)** and we compute the GDP using always the prices of that year

Real GDP (**at constant prices**) = value of current production based on prices of a year chosen as reference (**base year, b**):

$$Y_t = p_{1b} Q_{1t} + p_{2b} Q_{2t} + \dots$$

- It changes only if the produced quantity changes (different economy size) => stats cleaned from the price dynamics
- Choice of b is arbitrary
- When b is changed, figures must be updated the figures must be updated accordingly

Base year 2012

Year	Quantity of Cars	Price of Cars	Real GDP (in 2012 dollars)
2011	10	\$20	\$240
2012	12	\$24	\$288
2013	13	\$26	\$312

Prices stay
constant

- Real GDP in 2011 (in 2012 dollars) = 10 cars x \$24 per car = \$240
- Real GDP in 2012 (in 2012 dollars) = 12 cars x \$24 per car = \$288
- Real GDP in 2013 (in 2012 dollars) = 13 cars x \$24 per car = \$312

Changes reflect real
changes in the size of
the economy

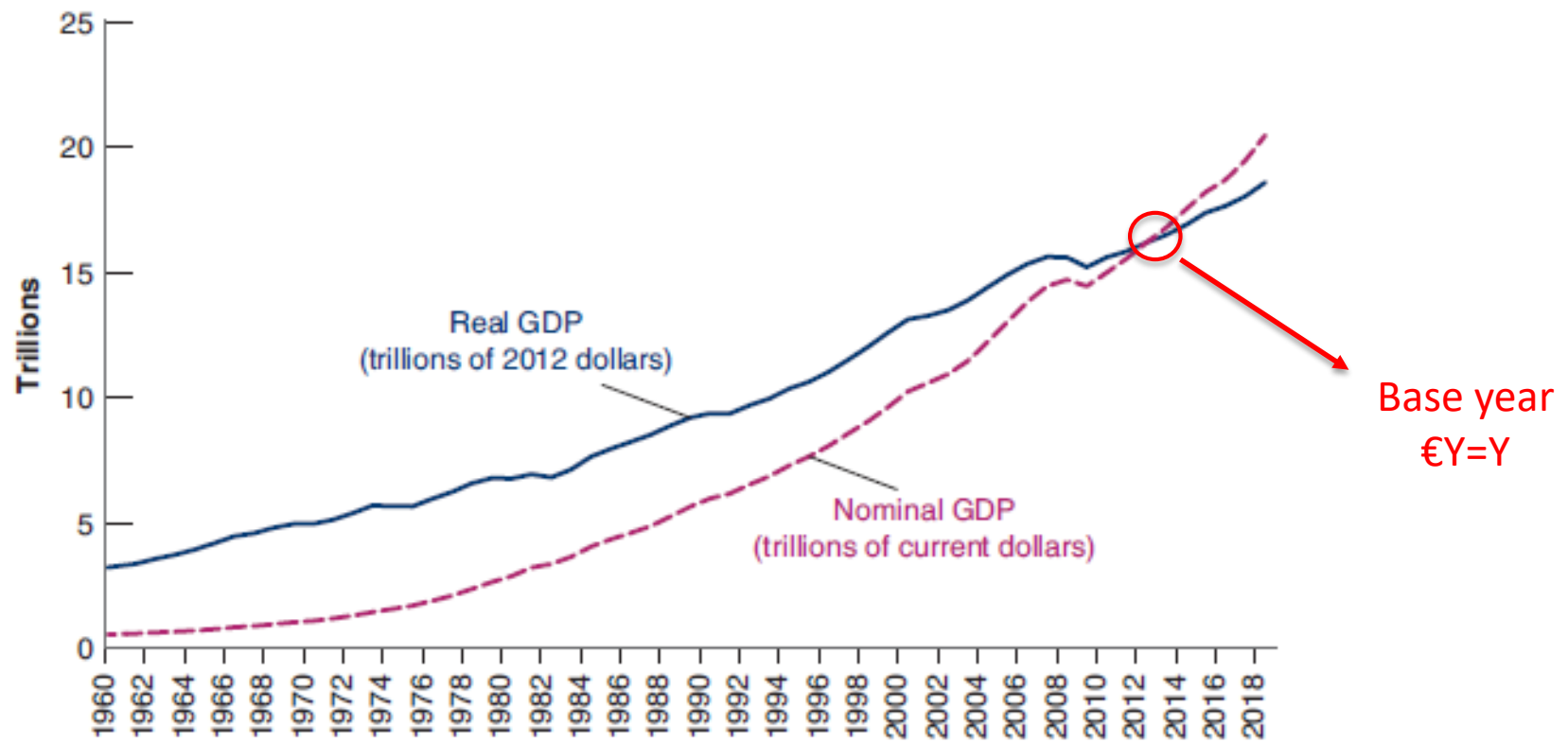
Base year 2011

Year	Quantity of Cars	Price of Cars	Real GDP (in 2012 dollars)
2011	10	\$20	\$240
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2013	13	\$26	\$312

- Real GDP in 2011 (in 2011 dollars) = 10 cars x \$20 per car = \$200
- Real GDP in 2012 (in 2011 dollars) = 12 cars x \$20 per car = \$240
- Real GDP in 2013 (in 2011 dollars) = 13 cars x \$20 per car = \$260

Changing b we obtain $\neq$ figures
but qualitatively they are the
same!

NOMINAL vs. REAL GDP



NOMINAL vs. REAL GDP

NOMINAL GDP

GDP at current prices

GDP in current €

REAL GDP

GDP at constant prices

GDP in terms of goods

GDP in constant €

GDP adjusted for inflation

GDP in chained (2012) €

GDP in 2012 €

In general, GDP “alone” refers to REAL GDP

HOW DO WE EVALUATE AN ECONOMY'S PERFORMANCE?

- ① *Dimension* → real GDP level (bill €, \$, £....) at time t
- Useful to compare countries considering also exchange rate and relative prices

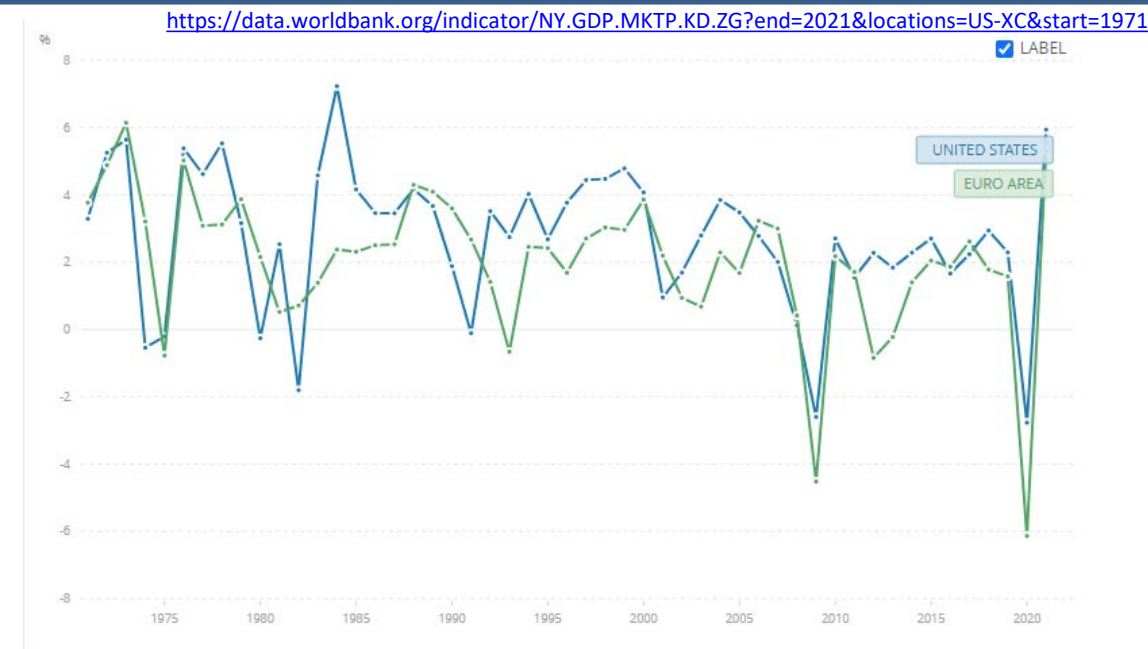
- ② *Evolution* → real GDP growth rate (%) in a given period, btw t-1 and t

- The rate of growth is the percentage change in GDP between period t -1 and period t

- Real GDP growth rate
$$g_y = \frac{(Y_t - Y_{t-1})}{Y_{t-1}}$$

- Nominal GDP growth rate
$$g_{ey} = \frac{(\epsilon Y_t - \epsilon Y_{t-1})}{\epsilon Y_{t-1}}$$

BUSINESS CYCLE



Annual percentage growth rate of GDP at market prices based on constant 2015 local currency expressed in U.S. dollars

The real GDP growth rate defines the [phases of the business cycles](#)

- ① **Expansions:** periods of **positive growth rate**
- ② **Recessions:** periods of **negative growth rate**
 - Statistical recession = *2 subsequent quarters* or more of negative growth
 - Clear out seasonal natural fluctuations

REAL GDP WITH MANY GOODS

What happens in the reality where we have many goods?

When many goods are produced

- GDP is a weighted average of the value of all final goods and service produced
- relative prices are the weights
 - If one good costs twice as much as another, that good counts for twice as much as the other

EXERCISE

- *Compute nominal GDP growth*

Year	Price of Steak	Quantity of Steak	Price of Basketballs	Quantity of Basketballs
2014	\$10	50	\$25	100
2015	\$12	50	\$28	100

Changes in Nominal GDP due to inflation

Year	Price of Steak	Quantity of Steak	Price of Basketballs	Quantity of Basketballs
2014	\$10	50	\$25	100
2015	\$12	50	\$28	100

- Nominal GDP in 2014 = $\text{€}Y_{14} = 10 * 50 + 25 * 100 = 3.000$
- Nominal GDP in 2015: $\text{€}Y_{15} = 12 * 50 + 28 * 100 = 3.400$
- GDP nominal change = \$ 400
- $g_{\text{€}Y} = (3400 - 3000) / 3000 = 0,13 = + 13.3\%$

Changes in prices and quantities

Compute nominal GDP growth

*Compute real GDP growth when the **base year is 2014***

Year	Price of Steak	Quantity of Steak	Price of Basketballs	Quantity of Basketballs
2014	\$10	50	\$25	100
2015	\$12	60	\$28	105

- Nominal/real GDP in 2014: $\text{€}Y_{14} = Y_{14} = 10 \cdot 50 + 25 \cdot 100 = 3.000$
- Nominal GDP in 2015: $\text{€}Y_{15} = 12 \cdot 60 + 28 \cdot 105 = 720 + 2940 = 3.660$
- GDP nominal change = \$660 $\Rightarrow g_{\text{€}y} = (3660 - 3000)/3000 = 0,22 = +22\%$
- Real GDP in 2015 (in 2014 prices): $Y_{15} = 10 \cdot 60 + 25 \cdot 105 = 600 + 2625 = 3.225$
- GDP real change = \$225 $\Rightarrow g_y = (3225 - 3000)/3000 = 0,075 = 0,075 = +7.5\%$

Changes in prices and quantities

Compute real GDP growth when the *base year is 2015*

Year	Price of Steak	Quantity of Steak	Price of Basketballs	Quantity of Basketballs
2014	\$10	50	\$25	100
2015	\$12	60	\$28	105

- Nominal GDP in 2014: $\text{€}Y_{14} = 10 \cdot 50 + 25 \cdot 100 = 3.000$
- Real GDP in 2014 (in 2015 prices): $Y_{14} = 12 \cdot 50 + 28 \cdot 100 = 600 + 2800 = 3.400$
- Nominal/real GDP in 2015: $\text{€}Y_{15} = Y_{15} = 12 \cdot 60 + 28 \cdot 105 = 720 + 2940 = 3.660$
- GDP real change = \$220 $\Rightarrow g_y = (3660 - 3400) / 3400 = 0,064 = + 6,4\%$

Euro Area Inflation Rate

<https://tradingeconomics.com/euro-area/inflation-cpi>



INFLATION (PRICE DYNAMICS)

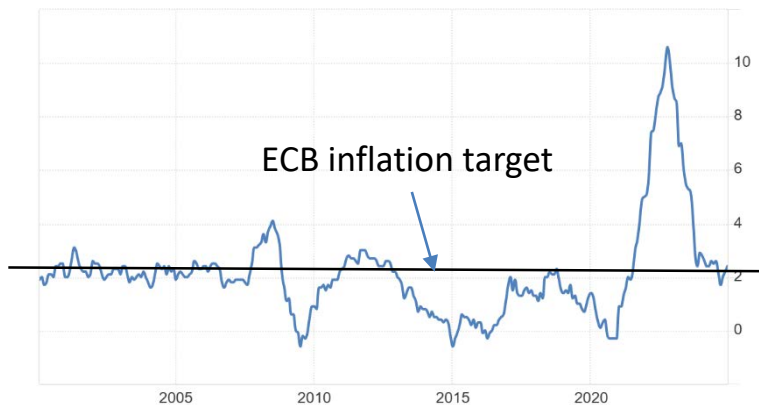
THE PRICE DYNAMICS

- In the economy prices *react* to changes in the D and/or S
 - Prices can be *flexible* o *sticky*
- *How do we call their variation?*

Inflation: a sustained rise in the general level of prices

Deflation: a sustained decline in the general level of prices(negative inflation)

EA Inflation Rate - percent



Source: tradingeconomics.com | EUROSTAT

US Inflation Rate - percent



Source: tradingeconomics.com | U.S. Bureau of Labor Statistics

THE INFLATION RATE

- The rate at which the price level changes => % change in the general level of prices between two periods
- *What prices do we consider?*
 - ① **Prices of the production** (not necessarily consumed)
 - GDP deflator – P_t
 - ② **Prices of consumed goods and services** (cost of living)
 - Consumer Price Index – CPI_t

① PRODUCTION PRICES

- IDEA: If nominal GDP increases faster than the real GDP, the difference must come from an increase in prices
 - GDP deflator = Ratio btw nominal and real GDP in a given year

$$P_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t} = \frac{\text{€}Y_t}{Y_t}$$

- It measures the average price of the final goods and services produced in the economy
 - Index number equal to 1 in the base year
- Its rate of change (%) represents the rate of inflation of the production

$$\pi_t^P = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

EXERCISE

- Compute the GDP deflator and the inflation rate using *2020 as base year*

Year	Wine		Pizza	
2020	200	4€	110	2€
2021	190	4,5€	100	3€
2022	300	6€	120	4€

$$\text{€}Y_{20} = Y_{2020} = 200 \times 4 + 110 \times 2 = 800 + 220 = 1020$$

$$P_{20} = \frac{1020}{1020} = 1$$

$$\text{€}Y_{21} = 190 \times 4,5 + 100 \times 3 = 855 + 300 = 1155$$

$$Y_{21} = 190 \times 4 + 100 \times 2 = 760 + 200 = 960$$

$$P_{21} = \frac{1155}{960} = 1,2$$

$$\text{€}Y_{22} = 300 \times 6 + 120 \times 4 = 1800 + 480 = 2280$$

$$\text{€}Y_{22} = 300 \times 4 + 120 \times 2 = 1200 + 240 = 1440$$

$$P_{22} = \frac{2280}{1440} = 1,6$$

$$\pi_{21}^P = \frac{1,2 - 1}{1} = 0,2$$

$$\pi_{2022}^P = \frac{1,6 - 1,2}{1,2} = 0,31$$

IS THE GDP DEFLATOR A GOOD PRICE INDEX?

It depends on the purpose since...

- Not all produced goods go to final national consumers
 - Some produced goods are
 - Not sold domestically (*exports*)
 - Not sold to final consumers (but to firms or to the government)
- Not all consumed goods are produced by domestic firms
 - Some of the goods bought by consumers produced abroad (*imports*)

② COST OF LIVING

- **CONSUMER PRICE INDEX** = it measures in the national currency (€, \$,...) the cost of a basket of goods and services consumed by the typical urban consumer
 - It measures the **average price** of the goods and services **consumed** in the economy
 - **Prices** are **weighted** according to the **purchase frequency**
 - It accounts for **changes** in the cost of living and the **purchasing power of wages**
 - Computed monthly
- Inflation rate based in the CPI = % change in the CPI
 - Monetary Policy in EU is set using the Harmonised Index of Consumer Price (HICP)
- Hyperinflation: extremely high % change in the CPI (100% in 24h)

Same procedure as before but considering **consumed goods** irrespective from where they are produced

1. Identify the consumer basket
2. Compute the CPI

CPI = Ratio btw nominal and real value of the consumed basket in a given year

$$CPI_t = \frac{\text{Nominal Basket Value}_t}{\text{Real Basket Value}_t}$$

3. Compute its **rate of change (%)** that is the rate of inflation of the consumed goods

$$\pi_t^{CPI} = \frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} = \frac{CPI_t}{CPI_{t-1}} - 1$$

EXERCISE

Consider the following economy where:

- bread and meat are produced
- bread and wine are consumed

	Bread		Meat		Wine	
2020	100	3€	100	12€	30	2€
2021	110	3,5€	110	12,5€	45	3€

Compute the rate of inflation using the GDP deflator and the CPI using as *base year 2020*

INFLATION WITH GDP DEFLATOR

- We do not have to consider wine since it is not produced domestically

$$P_{2020} = \frac{100 \times 3 + 100 \times 12}{100 \times 3 + 100 \times 12} = 1$$

$$P_{2021} = \frac{110 \times 3,5 + 110 \times 12,5}{110 \times 3 + 110 \times 12} = \frac{385 + 1375}{330 + 1320} = \frac{1760}{1650} = 1,06$$

$$\pi_{2021}^P = \frac{1,06 - 1}{1} = 0,06 = +6\%$$

INFLATION WITH CPI

- We do not have to consider meat since it is not consumed domestically

$$CPI_{2020} = \frac{100 \times 3 + 30 \times 2}{100 \times 3 + 30 \times 2} = 1$$

$$IPC_{2021} = \frac{110 \times 3,5 + 45 \times 3}{110 \times 3 + 45 \times 2} = \frac{385 + 135}{330 + 90} = \frac{520}{420} = 1,23$$

$$\pi_{2021}^{CPI} = \frac{1,23 - 1}{1} = 0,23$$



Istat

Istituto Nazionale
di Statistica

4 February 2025
Istat

PRESS RELEASE

CONSUMER PRICE INDICES. BASKET, WEIGHT STRUCTURE AND SURVEY UPDATES

As every year, Istat reviews the list of items included in the consumer price index basket and updates as well the survey techniques and the weights of each item to compile inflation.

The changes made in 2025, with reference to both the weights and the basket, reflect the constant evolution of household spending behaviours but also the impact of events which affect purchasing decisions and structure of consumer spending.

The 2025 basket for the Italian consumer price indices for the whole nation (NIC) and for blue and white-collar households (FOI) is made up of 1,923 elementary products (1,915 in 2024), which are grouped into 1,046 products and 424 product aggregates.

The 2025 basket for the Italian harmonized index of consumer prices (HICP) is made up of 1,944 elementary products (they were 1,936 in 2024), grouped into 1,065 products and 428 product aggregates.

The review of the basket of products takes into account the changes in the household spending patterns (also due to legal provisions) and enriches, representing consolidated consumption.

https://www.istat.it/wp-content/uploads/2025/02/Basket_2025.pdf

TABLE 2. CONSUMER PRICE INDICES BASKET 2025: NEW PRODUCTS

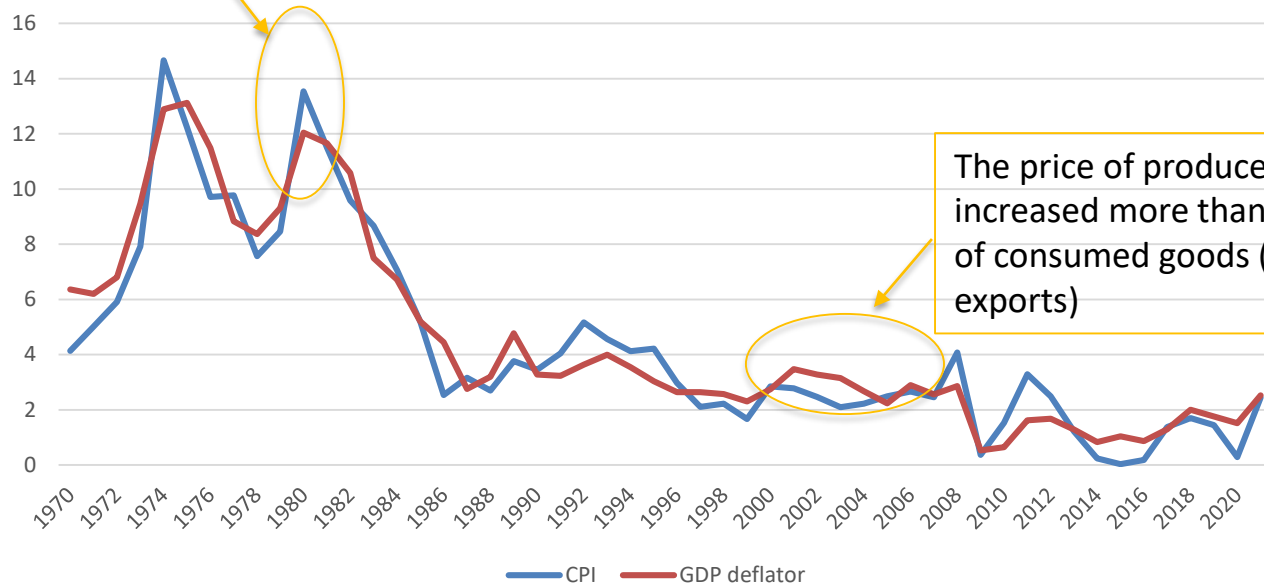
NEW

- Speck
- Women's shorts
- Ceiling lamp
- Mattress topper
- Bicycle inner tube
- Windshield Wiper Blades
- Dog waste bags
- Pet bowls
- Ice cream cone

INFLATION RATES

The price of consumed goods increased more than the price of produced goods (role of imports)

Inflation, Euro Area



The price of produced goods increased more than the price of consumed goods (role of exports)

- The CPI and GDP deflator move together most of the time => inflation rates, computed using either one or the other, are largely similar
- Some exceptions that reflect the price dynamics of import and exports

IS INFLATION AN ISSUE? WHY?

It depends!

- Symmetric vs. asymmetric inflation
- Uncertainty

PURE INFLATION

- *Pure inflation* is the proportional increase in all prices (including interest rates and wages) =>
 - Relative prices unaffected
 - Real wages unchanged
 - No impact on resources allocation

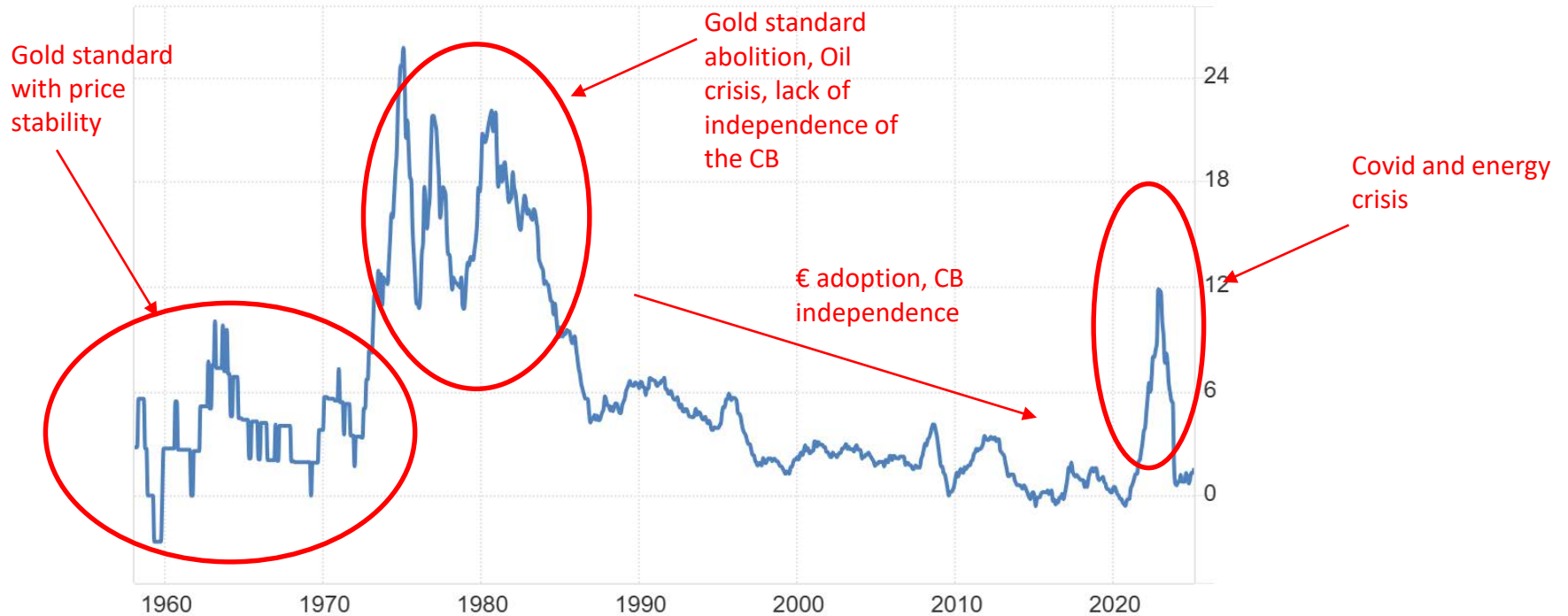
... but pure inflation does not exist, some prices are sticky

INFLATION COSTS

- *Why do economists care about inflation?*
- *It is costly for the economy since*
 - *Uncertainty* makes difficult to take optimal decisions
 - Consumption
 - Investment
 - *Distortions* related to changes in
 - relative prices => welfare loss!!
 - income *distribution* (retirees vs. workers)
- *What is the “best” rate of inflation?*
 - For most economists the rate of inflation must be low and stable, between 1 and 4%
 - ...but not zero (to leave space for the monetary policy)

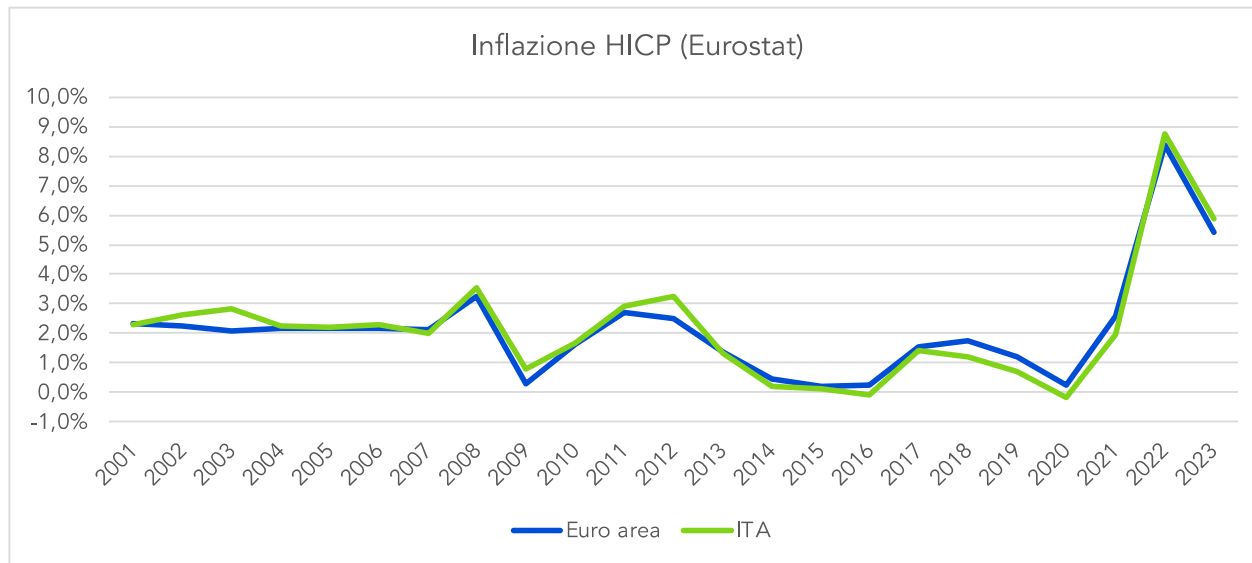
CASE STUDY: ITALY

IT Inflation Rate - percent



Source: tradingeconomics.com | National Institute of Statistics (ISTAT)

WE ARE NOT ALONE



GROWTH RATES AND INFLATION

$$P_t = \frac{\epsilon Y_t}{Y_t}$$

$$\Delta\% P_t = \Delta\% \frac{\epsilon Y_t}{Y_t}$$

...

$$\pi_t \approx g_{\epsilon Y_t} - g_{Y_t}$$

$\Leftrightarrow$

$$g_{\epsilon Y_t} \approx \pi_t + g_{Y_t}$$

Approximatively
equal

Rule: The rate of growth of nominal GDP is approximately equal to the rate of inflation plus the rate of growth of real GDP
 $\Leftrightarrow$
 the inflation rate computed with the GDP deflator is the difference between the rate of growth of the nominal and real GDP

QUICK CHECK

If the inflation rate computed with the GDP deflator is 3% and the inflation rate computed with CPI is 5%, :

- a) the prices of imported goods have risen more than the prices of domestically produced goods.
- b) the prices of domestically produced goods have risen faster than consumer goods prices.
- c) the inflation rate for services has been lower than that for goods.
- d) there is no difference between how the CPI and GDP deflator measure inflation.

QUICK CHECK

In country Macro, there are only two firms: a steel company and a car manufacturer. The steel company has revenues of 40, with 30 coming from the sale of steel to the car manufacturer and 10 from the sale of steel to final consumers. The car manufacturer uses steel as an intermediate good in the production of cars, which it sells to final consumers, generating revenues of 80. The country's GDP is:

- a) 80
- b) 50
- c) 90
- d) 70

A decrease in the nominal GDP by 3% can correspond approximately to

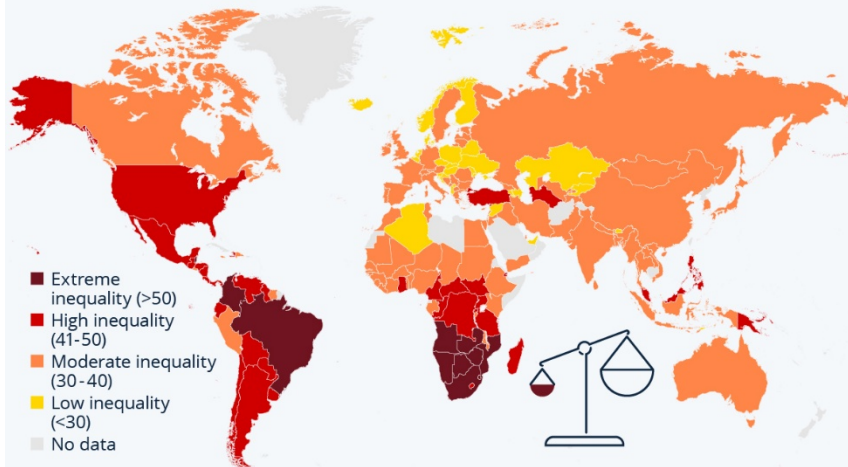
- a) an increase in real GDP by 6% and an increase in prices by 3%.
- b) a decrease in real GDP by 2% and an increase in prices by 5%.
- c) an increase in real GDP by 7% and a decrease in prices by 10%.
- d) a decrease in real GDP by 10% and a decrease in prices by 7%.

FINAL THOUGHTS

INEQUALITIES (not only in income)

The State of Global Income Inequality

Level of income inequality by country according to the Gini index (0=perfect equality, 100=total inequality)*



* Latest available Gini indexes per country

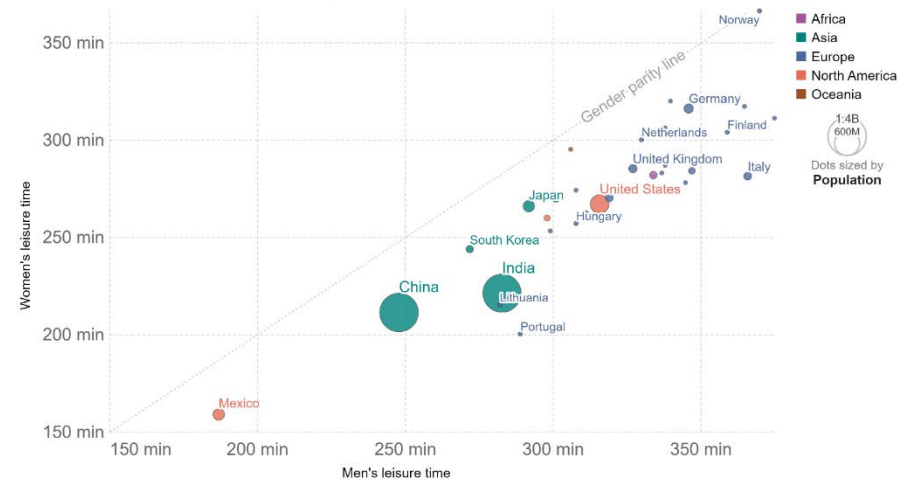
Source: The World Bank - Poverty and Inequality Platform (Sep. 2024)



statista

Gender gap in leisure time, 2020

Average minutes spent on leisure activities, per day, by sex (ages 15-65). For most countries surveys were conducted between 2009 and 2016, but surveys for some countries are older.



Source: OECD Gender Data

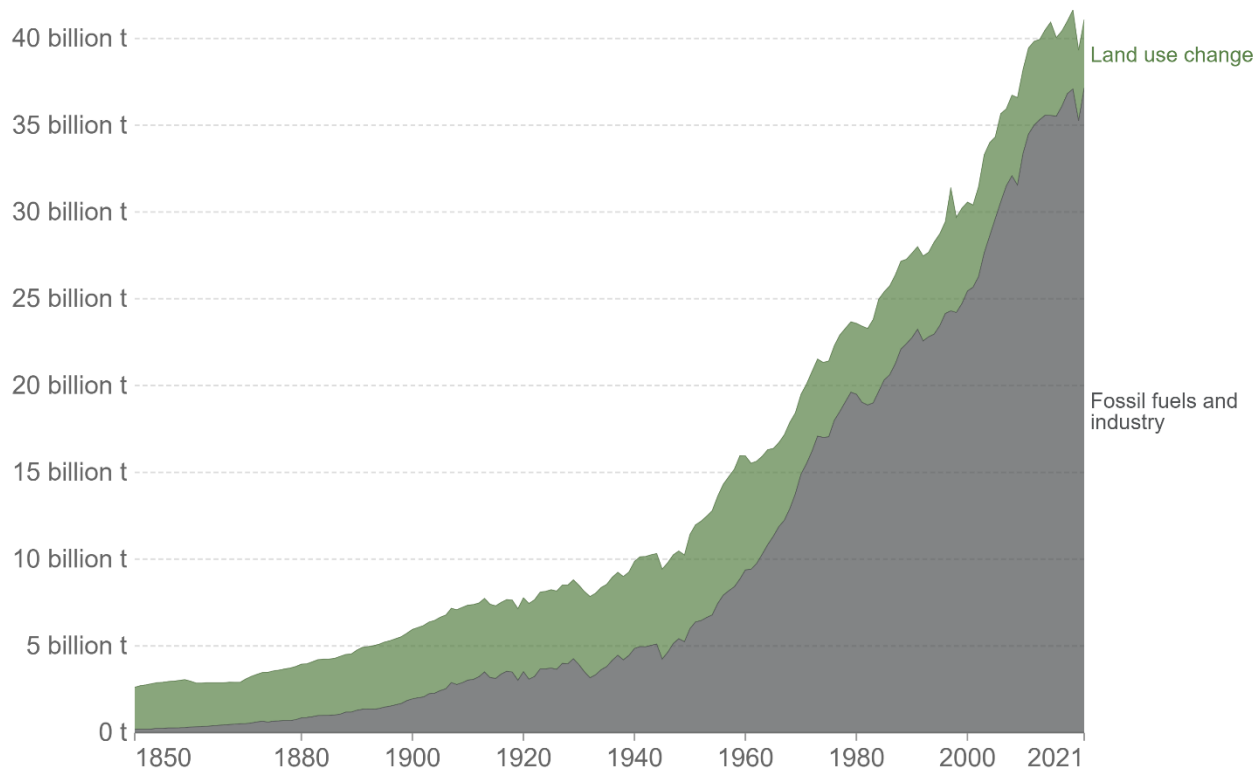
Note: Leisure activities include: sports, attending events, visiting friends, watching TV and other leisure activities.

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GDP GROWTH AND POLLUTION

Global CO₂ emissions from fossil fuels and land use change, World

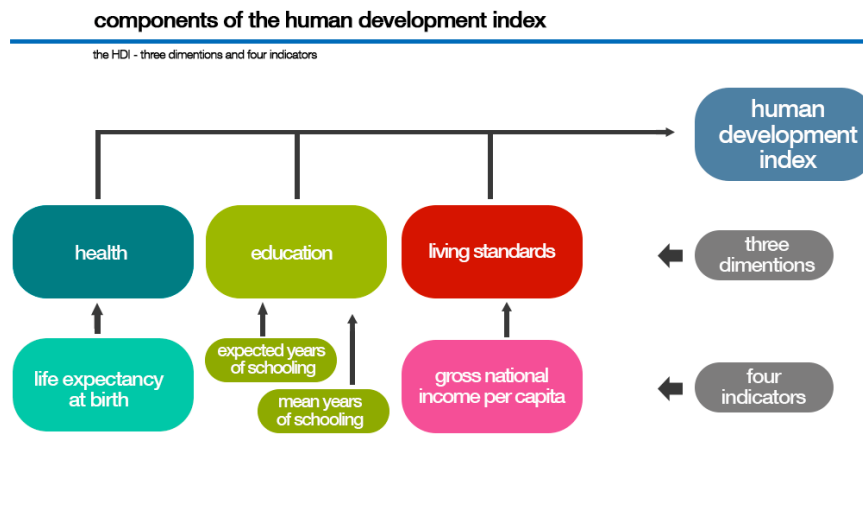
Our World
in Data



Source: Our World in Data based on the Global Carbon Project (2022) OurWorldInData.org/co2-and-other-greenhouse-gas-emissions • CC BY

SOME CORRECTIONS

- HDI



- Revision to the System of National Accounts (SNA)
 - Research & Development (R&D) purchases have been reclassified from intermediate consumption to investments, which increases GDP
 - Illegal activities included (estimates)

GDP AS A PROXY

Human Development Index vs. GDP per capita, 2022

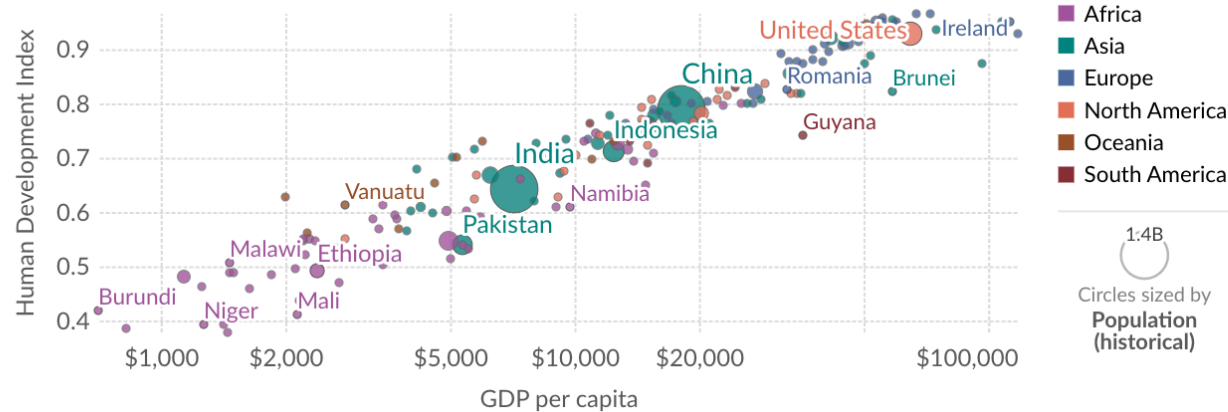
The Human Development Index (HDI) is a summary measure of key dimensions of human development: a long and healthy life, a good education, and a decent standard of living. GDP per capita is adjusted for inflation and differences in the cost of living between countries.

Our World
in Data

Table Chart

Select countries and regions

Settings



1990

2022

Data source: UNDP, Human Development Report (2024); World Bank (2023) – [Learn more about this data](#)

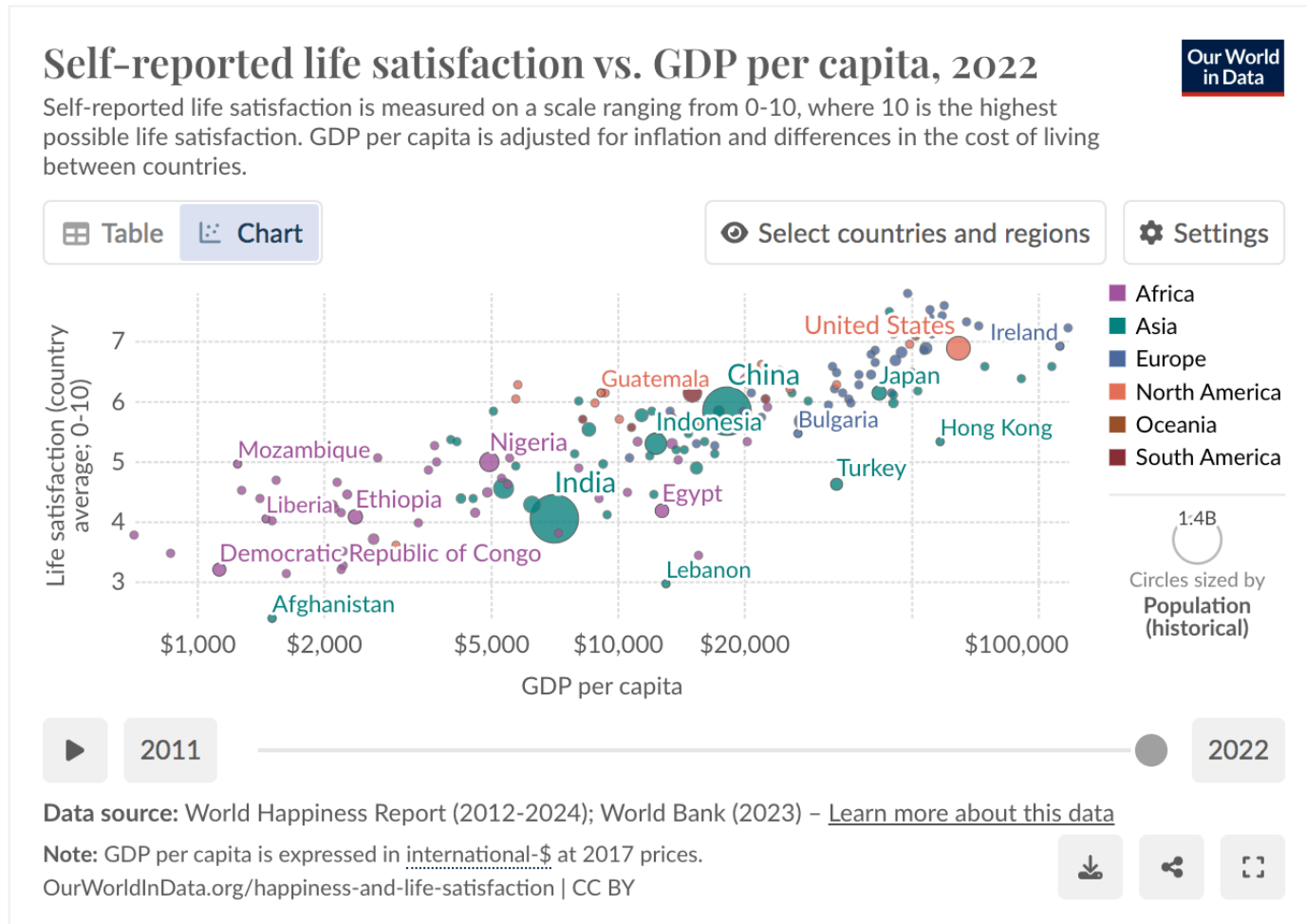
Note: GDP per capita is expressed in international-\$ at 2017 prices.

OurWorldInData.org/human-development-index | CC BY



<https://ourworldindata.org/grapher/human-development-index-vs-gdp-per-capita>

Money doesn't buy happiness, but...



<https://ourworldindata.org/grapher/gdp-vs-happiness>

Time for
questions!



TAKE AWAY AND KEY CONCEPTS

- GDP
 - Computation based on
 - Production
 - VA
 - Income
 - Expenditure
 - Nominal vs. Real GDP
 - GDP per capita
 - Growth rates: expansion, recession
 - Limitations and beyond GDP
- Inflation
 - GDP deflator
 - CPI